

TikTok Analysis Report

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February 18, 2026

Abstract

I run a TikTok account called `@humbletiker` where I post “goodnight challenge” videos. Up to this point, I’ve mostly improved by intuition and trial-and-error. In this project, I built a full pipeline to collect my post metrics and audio, transcribe the audio with Whisper, clean everything into a single analysis table, and then run a set of analyses to find actionable patterns. The goal is simple: identify what reliably correlates with better-performing videos (mostly higher views), and turn that into concrete changes I can implement.

1 Introduction

I’ve been creating video content on my TikTok account `@humbletiker` (pronounced “Humble Tokker”) for about 6 months now. I’ve grown to around 14k followers and over 1,000,000 total video views by making fun “goodnight challenge” videos. Note that I did *not* invent the goodnight challenge format, it’s a popular trend on TikTok that various creators do their own version of. The main idea of these challenge videos is that I say goodnight to a specific group, and only viewers who meet that criteria “win.”

Before this, I hadn’t used any serious data analysis to improve content performance. I mostly iterated based on surface level metrics, taking educated guesses at the factors that increased video performance. In this `tiktok-analysis` project, my goal is to extract meaningful insights that tell me what to lean into more. For performance, I mainly care about higher view counts (and I also look at related engagement metrics like likes and share rate).

2 Pre-processing and Data Collection

This project is basically: (1) collect the raw TikTok data, (2) turn the spoken audio into text, (3) clean and merge everything into one table I can analyze.

2.1 Scraping / Exporting Post Metrics

First I gathered post-level metrics for my videos. Conceptually, each post becomes one row in a dataset with fields like:

- video identifier (or URL / post ID)
- timestamp / post date
- views
- likes
- shares (or share rate)

- any other available metadata (duration, etc.)

The key thing is that I wanted a consistent schema so I could join this with transcript-based features later.

2.2 Downloading Audio and Transcribing with Whisper

Next, I downloaded the audio for each video and ran it through OpenAI Whisper to produce transcripts. The transcript is what unlocks the fun part of this project: instead of only analyzing what *format* a video is, I can analyze what I *actually say* and which phrases show up in high-performing videos.

For each video, I store:

- raw transcript text
- cleaned transcript text (normalized + stripped)

2.3 Cleaning + Merging Into One CSV

After transcription, I cleaned the data so it's usable for analysis:

- Standardized column names (so merges don't break).
- Removed obvious transcription noise (extra whitespace, repeated filler, etc.).
- Normalized text (lowercasing, consistent punctuation handling).
- Ensured numeric metrics are correctly typed (views as integer, rates as floats).
- Joined metrics and transcripts into a single table keyed by video.

The end result is one main CSV where each row is a video and columns include both performance metrics and transcript-derived features.

3 Analysis Approach (Without Results Yet)

Before getting into outcomes, here's how I approached analysis in a way that stays grounded and repeatable.

3.1 Define What “Success” Means

I treated **views** as the primary success metric, since that's the clearest top-of-funnel performance indicator. Then I also looked at supporting metrics like:

- likes (as a proxy for enjoyment)
- share rate (as a proxy for virality / “I want my friend to see this”)

3.2 Compare Content Formats

A big part of my content is template-driven. So I bucketed videos into format types, like:

- “goodnight to everyone except ...”
- “goodnight to no one except ...”

Then I compared average/median performance across buckets. I used both mean and median because TikTok performance is super skewed (one viral post can distort means).

3.3 Phrase-Level Signal From Transcripts

Once I had transcripts, I ran a phrase-level analysis:

- Identify recurring phrases / n-grams.
- Split videos into “top” and “bottom” performance groups.
- For each phrase, compare how often it appears in top vs bottom.

This gives me something actionable: if a phrase shows up a lot in top videos but never in bottom videos, that’s a strong clue that the line is doing something (hooking attention, prompting engagement, etc.). It’s not perfect causality, but it’s extremely useful for content iteration.

4 Takeaways From the Analysis

At this point, I *do* have takeaways from the analysis itself, but I don’t yet have the *follow-up experiment results* from implementing them. So I’m treating these as “best bets” backed by my data so far.

4.1 Takeaway 1: The “no one except” format dominates

When I compared the two main goodnight templates, the “**no one except**” format was way stronger than “**everyone except**”.

Format	<i>n</i>	Mean Views	Median Views	Mean Likes	Mean Share Rate	Median Share Rate
noone_except	30	71,215.7	4,444.5	290.7	0.0538	0.0577
everyone_except	15	2,374.9	1,393.0	14.3	0.0268	0.0386

What I’m going to do: I’m going to focus way more on the `noone_except` format. That means more variations, more hooks, and more testing within that template, because the baseline performance is just massively better in my dataset.

4.2 Takeaway 2: “share plus more” is a huge positive signal

The cleanest phrase-level insight I found is that when I do my little CTA about hitting “**share plus more**”, those videos correlate strongly with being top performers.

Here are the key numbers from the phrase analysis:

- Phrase “**plus**”: appears in 11 top videos and 0 bottom videos (top rate = 0.7857, bottom rate = 0.0000).
- Phrase “**share plus**”: appears in 11 top videos and 0 bottom videos (top rate = 0.7857, bottom rate = 0.0000).

That’s a super strong separation: it shows up in about 79% of top videos and *none* of the bottom ones.

What I’m going to do: I’m going to include that CTA in basically every video now, and I’m also going to test saying it **earlier** so more people actually hear it before swiping.

4.3 Takeaway 3: “birthday” looks promising but needs more data

Another phrase that popped as a potential positive signal is “**birthday**”, specifically:

- Phrase “**tonight birthday**”: appears in 4 top videos and 0 bottom videos (top rate = 0.2857, bottom rate = 0.0000).

The problem is the sample size. It’s only 4 top videos, so I don’t want to overclaim it’s a guaranteed win.

What I’m going to do: I’m still going to implement more birthday-related lines and scenarios, mainly to increase the sample size and see if it holds up consistently.

5 What’s Next (Implementation Results Pending)

Right now, this report covers what I found from analyzing my existing videos. The next phase is the part I’m most excited about: actually implementing these changes and measuring whether they causally improve performance.

My plan is:

- Post a batch of new videos emphasizing `noone_except`.
- Standardize the “share plus more” CTA, and test early vs late placement.
- Add more birthday content intentionally to validate whether it’s real or noise.

Once I have enough new posts, I’ll rerun the same analysis and write a follow-up section comparing *before vs after*.

End of report (for now).